

UNIT 3. Introduction to HTML

What is HTML?

HTML stands for **Hyper Text Markup Language** which is used for creating web pages and web applications. Let's see what is meant by Hypertext Markup Language, and Web page.

Hyper Text: HyperText simply means "Text within Text." A text has a link within it, is a hypertext. Whenever you click on a link which brings you to a new webpage, you have clicked on a hypertext. HyperText is a way to link two or more web pages (HTML documents) with each other.

Markup language: A markup language is a computer language that is used to apply layout and formatting agreements to a text document. Markup language makes text more interactive and dynamic. It can turn text into images, tables, links, etc.

Web Page: A web page is a document which is commonly written in HTML and translated by a web browser. A web page can be identified by entering an URL. A Web page can be of the static or dynamic type. With the help of HTML only, we can create static web pages.

Hence, HTML is a markup language which is used for creating attractive web pages with the help of styling, and which looks in a nice format on a web browser. An HTML document is made of many HTML tags and each HTML tag contains different content.

Simple example of HTML.

```
<!DOCTYPE>
<html>
<head>
<title>Web page title</title>
</head>
<body>
<h1>Write Your First Heading</h1>
<p>Write Your First Paragraph.</p>
</body>
</html>
```



History of HTML-

In the late 1980's , a physicist, Tim Berners-Lee who was a contractor at CERN, proposed a system for CERN researchers. In 1989, he wrote a memo proposing an internet based hypertext system.

Tim Berners-Lee is known as the father of HTML. The first available description of HTML was a document called "HTML Tags" proposed by Tim in late 1991. The latest version of HTML is HTML5, which we will learn later in this tutorial.

HTML Versions-

Since the time HTML was invented there are lots of HTML versions in market, the brief introduction about the HTML version is given below:

HTML 1.0: The first version of HTML was 1.0, which was the barebones version of HTML language, and it was released in 1991.

HTML 2.0: This was the next version which was released in 1995, and it was standard language version for website design. HTML 2.0 was able to support extra features such as form-based file upload, form elements such as text box, option button, etc.

HTML 3.2: HTML 3.2 version was published by W3C in early 1997. This version was capable of creating tables and providing support for extra options for form elements. It can also support a web page with complex mathematical equations. It became an official standard for any browser till January 1997. Today it is practically supported by most of the browsers.

HTML 4.01: HTML 4.01 version was released on December 1999, and it is a very stable version of HTML language. This version is the current official standard, and it provides added support for stylesheets (CSS) and scripting ability for various multimedia elements.

HTML5 : HTML5 is the newest version of HyperText Markup language. The first draft of this version was announced in January 2008. There are two major organizations one is W3C (World Wide Web Consortium), and another one is WHATWG(Web Hypertext Application Technology Working Group) which are involved in the development of HTML 5 version, and still, it is under development.

Features of HTML

- 1) It is a very **easy and simple language**. It can be easily understood and modified.
- 2) It is very easy to make an **effective presentation** with HTML because it has a lot of formatting tags.
- 3) It is a **markup language**, so it provides a flexible way to design web pages along with the text.
- 4) It facilitates programmers to add a **link** on the web pages (by html anchor tag), so it enhances the interest of browsing of the user.
- 5) It is **platform-independent** because it can be displayed on any platform like Windows, Linux, and Macintosh, etc.
- 6) It facilitates the programmer to add **Graphics, Videos, and Sound** to the web pages which makes it more attractive and interactive.
- 7) HTML is a case-insensitive language, which means we can use tags either in lower-case or upper-case.

HTML text Editors-

- An HTML file is a text file, so to create an HTML file we can use any text editors.
- Text editors are the programs which allow editing in a written text, hence to create a web page we need to write our code in some text editor.
- There are various types of text editors available which you can directly download, but for a beginner, the best text editor is Notepad (Windows) or TextEdit (Mac).
- After learning the basics, you can easily use other professional text editors which are, **Notepad++**, **Sublime Text**, **Vim**, etc.
- In our tutorial, we will use Notepad and sublime text editor. Following are some easy ways to create your first web page with Notepad, and sublime text.

Basic Structure of HTML Document-

Below a structure of an HTML document:

```
<html>
<head>
<title>Page title</title>
</head>
<body>
.
.
.
</body>
</html>
```

<!DOCTYPE>: It defines the document type or it instruct the browser about the version of HTML.

<html>: This tag informs the browser that it is an HTML document. Text between html tag describes the web document. It is a container for all other elements of HTML except <!DOCTYPE>

<head>: It should be the first element inside the <html> element, which contains the metadata (information about the document). It must be closed before the body tag opens.

<title>: As its name suggested, it is used to add title of that HTML page which appears at the top of the browser window. It must be placed inside the head tag and should close immediately.

<body>: Text between body tag describes the body content of the page that is visible to the end user. This tag contains the main content of the HTML document.

Attributes of body tag-

- background: It contains the URL of background image. It is used to set the background image.
- bgcolor: It is used to specify the background color of an image.
- alink: It is used to specify the color of active link.
- link: It is used to specify the color of visited links.

- text: It specify the color of text in a document.
- vlink: It specify the color of visited links.

How to Createa HTML Document-

Follow the four steps below to create your first web page with Notepad.

Step 1: Open Notepad (Windows)

Open the Notepad

Step 2: Create a New Document

Go to Notepad Menu: File > New

A New blank document will be opened and you can start writing your first HTML Program here.

Step 3: Write Some HTML code or Program

Write some HTML code.

Write your own HTML code or simply copy the following HTML Simple Program into notepad document.

```
<!DOCTYPE html>
<html>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
```

Step 4: Save the HTML Page

Go to Notepad Menu: File > Save (or use short-key CTRL + S)

It will ask you to Save the file on your computer. Give it a name with .html extension and Save it (for example program.html)

Note: HTML page should be saved with .html extension carefully.

Step 5: View the HTML Page using Browser

Web browsers are programs or software that are used to view Webpages/Websites.

You can find Internet Explored by default if using Windows Computer machine. You can also download other popular web browsers such as Google Chrome or Firefox. Use any of them.

Now Simply, open the saved HTML file in any browser:

Double click on the file or right-click on the file and choose "Open with" option to select other browser.

You HTML File will be opened in web browser and it will show output based on your html program.



What is Tag and attribute-

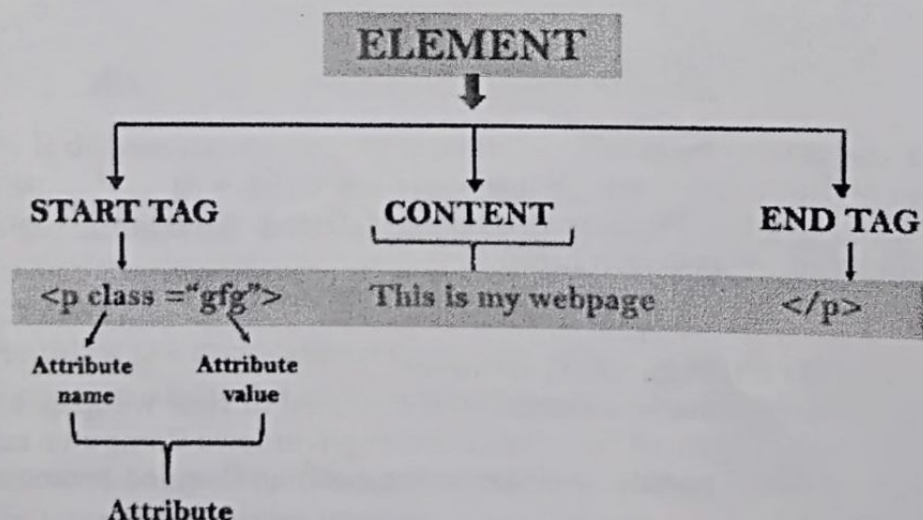
HTML Tags

- Tags are instructions that are embedded directly into the text of a HTML document. Each HTML tag describes that the browser should do something instead of simply displaying the text. In HTML, the tags begin with (<) and end with (>)
- HTML tags can be of two types.
- **Attribute:** An attribute in HTML provides extra information about the element, and it is applied within the start tag. An HTML attribute contains two fields: name & value.

Syntax

`<tag name attribute_name= " attr_value"> content </ tag name>`

- **Elements:** An HTML element is an individual component of an HTML file. In an HTML file, everything written within tags are termed as HTML elements.



Comments in HTML-

They begin with the `<!--` (starting comment tag) and ends with the close comment tag, i.e. `-->`. Comments in HTML can be visible if and only if anyone tries to view the source code of the page, but anything is written within `<!-- .... -->` does not get rendered when you will render your HTML based web page through your browser.

Syntax:


```
<!-- This is a commented code -->
```

Example:

```
<html>
<!-- Body of webpage starts from here -->
<body>
<h2>This is HTML comment example</h2>
</body>
</html>
```

HTML Headings-

A HTML heading or HTML h tag can be defined as a title or a subtitle which you want to display on the webpage. When you place the text within the heading tags `<h1>.....</h1>`, it is displayed on the browser in the bold format and size of the text depends on the number of heading.

There are six different HTML headings which are defined with the `<h1>` to `<h6>` tags, from highest level h1 (main heading) to the least level h6 (least important heading).

h1 is the largest heading tag and h6 is the smallest one. So h1 is used for most important heading and h6 is used for least important.

See this example:

```
<body>
<h1>Heading no. 1</h1>
<h2>Heading no. 2</h2>
<h3>Heading no. 3</h3>
<h4>Heading no. 4</h4>
<h5>Heading no. 5</h5>
<h6>Heading no. 6</h6>
</body>
```

Output-

Heading no. 1

Heading no. 2

Heading no. 3

Heading no. 4

Heading no. 5

Heading no. 6

HTML Paragraph-

HTML paragraph or HTML p tag is used to define a paragraph in a webpage. Let's take a simple example to see how it work. It is a notable point that a browser itself add an empty line before and after a paragraph. An HTML <p> tag indicates starting of new paragraph.

See this example:

```
<p>This is first paragraph.</p>
```

```
<p>This is second paragraph.</p>
```

```
<p>This is third paragraph.</p>
```

Output-

This is first paragraph.

This is second paragraph.

This is third paragraph.

Unclosed HTML Tags

Some HTML tags are not closed, for example br and hr.

**
 Tag:**br stands for break line, it breaks the line of the code.

<hr> Tag:hr stands for Horizontal Rule. This tag is used to put a line across the webpage.

1) Line Break (
)tag-

The
 tag inserts a single line break.

The
 tag is an empty tag which means that it has no end tag.

Example-

```
<body>
```

This text contains
a line break.

```
</body>
```

Output-

This text contains
a line break.

2) <hr> tag

HTML <hr> tag is used to specify a paragraph-level thematic break in HTML document. It is used when you abruptly change your topic in your HTML document. It draw a horizontal line between them. It is also called a Horizontal Rule in HTML.

Tag Attributes: The table given below describe the <hr> tag attributes:

ATTRIBUTE	VALUE	DESCRIPTION
Align	left center right	Used to specify the alignment of the horizontal rule.
noshade	noshade	Used to specify the bar without shading effect.
size	pixels	Used to specify the height of the horizontal rule.
width	pixels	Used to specify the width of the horizontal rule.

Example-

```
<body>
<p>HTML is a language for describing web pages.....</p>
<hr align="center" size=5 width=1000>
<p>CSS defines how to display HTML elements.....</p>
</body>
```

Output-

HTML is a language for describing web pages.

CSS defines how to display HTML elements.

<Pre> Tag:

The pre-formatted text tag is used to preserve the tabs, indentations, and linespacing in your original source text. Netscape and other graphical browsers display preformatted text in fixed font, such as Courier; other text appears is a proportional font, such as Geneva or Times.

Example:

```
<pre>
Happy New Year
All the Best.
</pre>
```